OOP MCQ Question Bank

60 questions | Difficulty 0.5 to 10 (step 0.5, 3 questions per level) | 4 options each

Q1. [Difficulty: 0.5] Which of the following is NOT a pillar of object-oriented programming?

A) Encapsulation
B) Inheritance
C) Compilation
D) Polymorphism

Answer: C

Q2. [Difficulty: 0.5] What is an object in OOP?

A) A primitive data type
B) An instance of a class
C) A function that returns a value
D) A blueprint for creating classes

Answer: B

Q3. [Difficulty: 0.5] What is a class in OOP?

A) A blueprint or template for creating objects
B) A type of loop
C) A variable that stores a number
D) A running instance of a program

Answer: A

Q4. [Difficulty: 1] In Java, which keyword is used to create a new object?

A) create
B) object
C) make
D) new

Answer: D

Q5. [Difficulty: 1] What does encapsulation mean?

A) Bundling data and the methods that work on it, and restricting direct access to the data
B) Creating many methods with the same name
C) Hiding the class name from the compiler
D) Acquiring properties from a parent class

Answer: A

Q6. [Difficulty: 1] What is inheritance?

A) A class containing objects of another class
B) A class acquiring the properties and methods of another class
C) Defining a method more than once
D) Creating several objects of the same class

Answer: B

Q7. [Difficulty: 1.5] What is a constructor?

A) A method that deletes an object
B) A method that can only be static
C) A special method that is called when an object is created
D) A method that must be called manually after creation

Answer: C

Q8. [Difficulty: 1.5] Which access modifier makes a member accessible only inside its own class?

A) protected
B) default (package-private)
C) public
D) private

Answer: D

Q9. [Difficulty: 1.5] What does polymorphism mean?

A) The same interface can show different behaviour for different objects
B) Storing many values in one variable
C) Creating a class from another class
D) Hiding data inside a class

Answer: A

Q10. [Difficulty: 2] What is method overloading?

A) Hiding a method from other classes
B) Calling a method from inside itself
C) Defining several methods with the same name but different parameters in the same class
D) Redefining a parent class method in a child class

Answer: C

Q11. [Difficulty: 2] What is method overriding?

A) Making a method private
B) Calling two methods at once
C) A subclass providing its own implementation of a method already defined in its parent class
D) Defining the same method name with different parameters

Answer: C

Q12. [Difficulty: 2] What is abstraction?

A) Hiding implementation details and showing only the essential features
B) Combining data and methods in a single unit
C) Making all members public
D) Copying one object into another

Answer: A

Q13. [Difficulty: 2.5] Which of the following is an example of compile-time polymorphism?

A) Method overloading
B) Virtual functions
C) Method overriding
D) Interface implementation

Answer: A

Q14. [Difficulty: 2.5] Which of the following is an example of run-time polymorphism?

A) Operator overloading
B) Constructor overloading
C) Method overloading
D) Method overriding

Answer: D

Q15. [Difficulty: 2.5] What is an abstract class?

A) A class with only static methods
B) A class that cannot have any methods
C) A class that cannot be instantiated directly and may contain abstract methods
D) A class that can only be used once

Answer: C

Q16. [Difficulty: 3] Which is true about interfaces and classes in Java?

A) A class can extend multiple classes
B) A class can implement multiple interfaces but extend only one class
C) An interface can extend only a class
D) A class can implement only one interface

Answer: B

Q17. [Difficulty: 3] What does the 'this' keyword refer to?

A) The parent class
B) The current object
C) The class's static members
D) The most recently created object

Answer: B

Q18. [Difficulty: 3] What does a 'static' member of a class mean?

A) It belongs to the class itself, not to any particular object
B) It is accessible only inside the class
C) It is created every time an object is created
D) It cannot be changed after it is set

Answer: A

Q19. [Difficulty: 3.5] What is the return type of a constructor?

A) int
B) The class type
C) A constructor has no return type
D) void

Answer: C

Q20. [Difficulty: 3.5] What is the diamond problem?

A) An infinite loop in constructors
B) Ambiguity when a class inherits from two classes that both inherit from the same base class
C) Two classes having the same name
D) A class having four constructors

Answer: B

Q21. [Difficulty: 3.5] Does Java support multiple inheritance of classes?

A) Only inside the same package
B) Yes, using the extends keyword twice
C) Yes, but only for abstract classes
D) No, only through interfaces

Answer: D

Q22. [Difficulty: 4] What is the difference between composition and inheritance?

A) Composition models 'is-a'; inheritance models 'has-a'
B) Composition is only for interfaces; inheritance is only for classes
C) There is no difference
D) Composition models a 'has-a' relationship; inheritance models an 'is-a' relationship

Answer: D

Q23. [Difficulty: 4] What is the purpose of a copy constructor?

A) To delete an existing object
B) To copy a class definition
C) To convert an object into a string
D) To create a new object as a copy of an existing object

Answer: D

Q24. [Difficulty: 4] In C++, when is a destructor called for a stack-allocated object?

A) When the object goes out of scope
B) Only when the program crashes
C) When the program starts
D) Only when delete is called

Answer: A

Q25. [Difficulty: 4.5] What is the difference between shallow copy and deep copy?

A) A shallow copy copies references of nested objects; a deep copy also copies the nested objects
B) A deep copy is always faster
C) A shallow copy copies only primitive fields; a deep copy copies only references
D) They are the same thing

Answer: A

Q26. [Difficulty: 4.5] What is a virtual function in C++?

A) A function that exists only in memory
B) A function with no body that must be inline
C) A function that cannot be overridden
D) A function whose call is resolved at run time based on the actual object type

Answer: D

Q27. [Difficulty: 4.5] What is the difference between == and equals() for objects in Java?

A) == compares content; equals() compares references
B) Both always compare content
C) == compares references; equals() compares content when it is overridden
D) Both always compare references

Answer: C

Q28. [Difficulty: 5] What does the 'S' in SOLID stand for?

A) Static Binding Principle
B) Single Responsibility Principle
C) Singleton Principle
D) Separation of Interfaces

Answer: B

Q29. [Difficulty: 5] What happens when a Java class is declared final?

A) Its methods cannot be called
B) It cannot be extended
C) It must be abstract
D) Its objects cannot be created

Answer: B

Q30. [Difficulty: 5] Can a static method be overridden in Java?

A) Yes, but only if it is public
B) Yes, like any instance method
C) Only if the class is abstract
D) No; a static method in a subclass hides the parent method instead

Answer: D

Q31. [Difficulty: 5.5] What does the Liskov Substitution Principle state?

A) A class should have only one reason to change
B) High-level modules should not depend on low-level modules
C) Classes should be open for extension but closed for modification
D) Objects of a subclass should be usable wherever the parent class is expected without breaking correctness

Answer: D

Q32. [Difficulty: 5.5] Can an abstract class in Java have a constructor?

A) No, abstract classes cannot have constructors
B) Yes, but it can never be called
C) Only if it is declared static
D) Yes; it runs when a subclass object is created

Answer: D

Q33. [Difficulty: 5.5] If a Java class defines no constructor, what happens?

A) The compiler provides a default no-argument constructor
B) The class uses its parent class's constructor directly
C) A compile-time error occurs
D) The class cannot be instantiated

Answer: A

Q34. [Difficulty: 6] What does the Open/Closed Principle state?

A) Software entities should be open for extension but closed for modification
B) Code should never be changed after release
C) Every class must be open to all other classes
D) Classes should always be declared final

Answer: A

Q35. [Difficulty: 6] In inheritance, in which order are constructors called?

A) Derived class constructor first, then base class
B) Only the derived class constructor is called
C) The order is random
D) Base class constructor first, then derived class constructor

Answer: D

Q36. [Difficulty: 6] Why would a class have a private constructor?

A) To make the constructor run faster
B) To allow the class to be inherited by anyone
C) To prevent instantiation from outside the class, as in the Singleton pattern
D) To make the class abstract

Answer: C

Q37. [Difficulty: 6.5] What does the Dependency Inversion Principle state?

A) Low-level modules should control high-level modules
B) High-level modules should depend on abstractions, not on concrete low-level classes
C) Classes should be created using the new keyword only
D) Dependencies should be removed from all classes

Answer: B

Q38. [Difficulty: 6.5] What is object slicing in C++?

A) Casting a base pointer to a derived pointer
B) Splitting a large object into smaller objects
C) Copying a derived class object into a base class object by value, losing the derived part
D) Deleting part of an object's memory

Answer: C

Q39. [Difficulty: 6.5] Why should the destructor of a polymorphic base class in C++ be virtual?

A) So the destructor can take parameters
B) So the derived class destructor runs when an object is deleted through a base pointer
C) So the base class cannot be inherited
D) So the destructor runs faster

Answer: B

Q40. [Difficulty: 7] What does the Interface Segregation Principle state?

A) Clients should not be forced to depend on methods they do not use
B) A class must hide all its interfaces
C) Interfaces must be implemented by exactly one class
D) All interfaces must have exactly one method

Answer: A

Q41. [Difficulty: 7] What is a covariant return type?

A) A return type that must exactly match the parent's
B) A method that returns different types on every call
C) A method that returns two values
D) An overriding method may return a subtype of the return type declared in the parent method

Answer: D

Q42. [Difficulty: 7] What is the difference between aggregation and composition?

A) They are identical
B) In composition the part cannot outlive the whole; in aggregation it can
C) Composition is used only with interfaces
D) In aggregation the part cannot outlive the whole; in composition it can

Answer: B

Q43. [Difficulty: 7.5] Can a Java interface contain method implementations?

A) Yes, but only private fields
B) Only if it is declared abstract
C) No, never
D) Yes, default and static methods (since Java 8)

Answer: D

Q44. [Difficulty: 7.5] A subclass overrides a public Java method but makes it private. What happens?

A) A run-time exception is thrown when it is called
B) A compile-time error, because the access level cannot be reduced
C) It works, and the method becomes private
D) It works but a warning is shown

Answer: B

Q45. [Difficulty: 7.5] What does the Law of Demeter suggest?

A) A method should talk only to its immediate collaborators, not to objects reached through chains of calls
B) Every class should have only one method
C) All data members should be public
D) Methods should never call other methods

Answer: A

Q46. [Difficulty: 8] In Java, are fields polymorphic like overridden methods?

A) Yes, they are resolved at run time by the object type
B) Only for static fields
C) Yes, but only for private fields
D) No; field access is resolved at compile time by the reference type

Answer: D

Q47. [Difficulty: 8] What is the Rule of Three in C++?

A) A function can have at most three overloads
B) If a class needs a custom destructor, copy constructor or copy assignment operator, it likely needs all three
C) A class may have at most three constructors
D) A class can inherit from at most three classes

Answer: B

Q48. [Difficulty: 8] What is the fragile base class problem?

A) A base class that cannot be inherited
B) Changes in a base class can unintentionally break its derived classes
C) A base class with too many methods
D) A base class that has no constructor

Answer: B

Q49. [Difficulty: 8.5] What is the risk of calling an overridable method from a constructor in Java?

A) The method can never be overridden
B) A compile-time error occurs
C) The constructor will run twice
D) The overridden method may run before the subclass fields are initialized

Answer: D

Q50. [Difficulty: 8.5] What is a vtable (virtual table)?

A) A table of function pointers used for dynamic dispatch of virtual functions
B) A table of exceptions thrown at run time
C) A table of all objects created in a program
D) A table of variable names used by the compiler

Answer: A

Q51. [Difficulty: 8.5] What does the 'explicit' keyword do on a C++ constructor?

A) Makes the constructor virtual
B) Prevents the compiler from using it for implicit conversions
C) Makes the constructor run before main
D) Prevents the class from being inherited

Answer: B

Q52. [Difficulty: 9] Why is making Square a subclass of Rectangle a classic Liskov violation?

A) A square has fewer sides than a rectangle
B) A square cannot have a constructor
C) Setting width and height independently breaks the invariant that a square's sides are equal
D) Rectangle is an abstract class

Answer: C

Q53. [Difficulty: 9] What is the Rule of Five in C++ (an extension of the Rule of Three)?

A) A class should have five virtual functions
B) It adds the move constructor and move assignment operator to the three special functions
C) A class can have at most five data members
D) A function should have at most five parameters

Answer: B

Q54. [Difficulty: 9] In Java, what rule applies when an overriding method declares checked exceptions?

A) It must throw every exception the parent throws
B) It may throw any checked exception
C) It cannot throw new or broader checked exceptions than the overridden method
D) It cannot throw any exception at all

Answer: C

Q55. [Difficulty: 9.5] How does virtual inheritance in C++ solve the diamond problem?

A) It forbids multiple inheritance
B) It makes the base class private
C) It ensures only one shared copy of the common base class exists in the derived object
D) It copies the base class twice

Answer: C

Q56. [Difficulty: 9.5] Which algorithm does Python use to compute the method resolution order (MRO) in multiple inheritance?

A) C3 linearization
B) Dijkstra's algorithm
C) Random resolution
D) Depth-first search

Answer: A

Q57. [Difficulty: 9.5] Under Design by Contract, what is required of an overriding method in a subtype?

A) It must keep both preconditions and postconditions unchanged in text
B) It may change both freely
C) It must not strengthen preconditions and must not weaken postconditions
D) It must strengthen preconditions and weaken postconditions

Answer: C

Q58. [Difficulty: 10] For overriding methods, which parameter type change is theoretically safe according to the Liskov principle?

A) A more general (contravariant) parameter type
B) Any type, as long as the name is the same
C) A more specific (covariant) parameter type
D) An unrelated parameter type

Answer: A

Q59. [Difficulty: 10] In C++, what happens when a base class constructor calls a virtual function that a derived class overrides?

A) A compile-time error occurs
B) The program always crashes
C) The base class version is called, because the derived part is not constructed yet
D) The derived class version is called

Answer: C

Q60. [Difficulty: 10] The Visitor design pattern is mainly used to emulate:

A) Multiple inheritance
B) Operator overloading
C) Double dispatch
D) Lazy initialization

Answer: C